SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Sreesha B 192126O034 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sreesha

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	2	2	2	2
Debugging	25%	1	1	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

Q5 10marks
1
1
1
1
1

2b/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

2. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails. Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

- A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

•Identifying the matching routing entry using the longest-prefix match.
•Determining the next-hop router selected for packet forwarding.
•Verifying whether the destination IP belongs to the intended subnet.
•Identifying any routing configuration errors.
•Optimizing the routing table to improve packet forwarding efficiency.
•Identifying the route that will be used if no matching destination exists.

Q1. Debugging IPv4 Addressing and Subnet Configuration

1. Problem identification

The workstation is configured with:

•IP address: `192.168.10.65`
•Subnet mask: `255.255.255.192` (`/26`)
•Default gateway: `192.168.10.1`

The workstation cannot communicate with devices in the network `192.168.10.0/26`.

The likely cause is an incorrect IP subnet assignment and an invalid default gateway for the configured subnet.

2.Calculate the subnet details

A `/26` subnet mask contains 26 network bits and 6 host bits.

- Subnet mask: `255.255.255.192`
- Block size: 256-192=64
- Subnet ranges in the last octet: 0–63, 64–127, 128–191, and 192–255.

The IP address `192.168.10.65` belongs to the subnet from 64 to 127.

Parameter	Result
Configured IP	`192.168.10.65`
Subnet mask	`255.255.255.192 (/26)`
Network address	`192.168.10.64`
Broadcast address	`192.168.10.127`
Valid host range	`192.168.10.65 – 192.168.10.126`
Total addresses	64
Usable host addresses	62

3. Verify the IP address and default gateway

The IP address `192.168.10.65` is valid for the subnet `192.168.10.64/26`, because it falls within the usable host range.

However, the default gateway `192.168.10.1` belongs to the subnet `192.168.10.0/26`, whose usable host range is `192.168.10.1–192.168.10.62`.

Therefore, the configured gateway is outside the workstation's subnet. In a conventional IPv4 LAN, the default gateway must be reachable on the local link.

The configuration is inconsistent with the intended network `192.168.10.0/26`.

4. Debugging and root cause

Identified issue	Explanation
Incorrect subnet membership	The workstation is in `192.168.10.64/26`, not `192.168.10.0/26`.
Invalid gateway for this configuration	`192.168.10.1` belongs to a different `/26` subnet.
Communication failure	The workstation cannot use the configured gateway as its normal on-link next hop.
Network design mismatch	The host address does not match the intended `192.168.10.0/26` network.

5. Optimized configuration

To place the workstation in the intended `192.168.10.0/26` subnet, configure it as follows.

Parameter	Corrected setting
IP address	`192.168.10.10`
Subnet mask	`255.255.255.192`
Default gateway	`192.168.10.1`
Network address	`192.168.10.0`
Broadcast address	`192.168.10.63`
Valid host range	`192.168.10.1–192.168.10.62`

The suggested host address `192.168.10.10` is an example and must be checked to ensure it is not already assigned to another device. The gateway must also be configured on the router interface connected to this subnet.

6. Usable host calculation

The formula for a conventional IPv4 subnet is:

Usable hosts=2^h-2
where h is the number of host bits. For `/26`:

h=32-26=6
Usable hosts=2^6-2=64-2=62

Final conclusion: The original IP address belongs to `192.168.10.64/26`, while the intended network and gateway belong to `192.168.10.0/26`. Assigning a valid host address such as `192.168.10.10`, retaining the `/26` mask, and using the on-link gateway `192.168.10.1` corrects the subnet mismatch. The corrected subnet supports 62 usable IPv4 host addresses.

Q2. Debugging IPv6 Addressing and Prefix Configuration

1. Problem identification

The two devices have the following IPv6 configurations:

•Device A: `2001:db8::ff00:42:8329/64`
•Device B: `2001:db8:0:0:1::1/64`

Both devices are connected to the same network, but communication fails. The first step is to expand the addresses and compare their network prefixes.

2.Expand both IPv6 addresses

IPv6 addresses contain eight groups of four hexadecimal digits. The double colon (`::`) represents one or more consecutive groups of zeros.

Device A Compressed address:

`2001:db8::ff00:42:8329/64`

Fully expanded address:

`2001:0db8:0000:0000:0000:ff00:0042:8329/64`
Device B Compressed address:

`2001:db8:0:0:1::1/64`

Fully expanded address:

`2001:0db8:0000:0000:0001:0000:0000:0001/64`

2.Determine the network prefixes

A `/64` prefix means the first four 16-bit groups (64 bits) represent the network prefix.

Parameter	Device A	Device B
IPv6 address	`2001:db8::ff00:42:8329`	`2001:db8:0:0:1::1`
Prefix length	`/64`	`/64`
Network prefix	`2001:db8:0:0::/64`	`2001:db8:0:0::/64`
Same subnet?	Yes	Yes

Important finding: Both addresses actually belong to the same `/64` subnet. The fifth group in Device B's address is part of its interface identifier, not its network prefix.

4. Debugging and root-cause analysis

The given addresses do not contain a subnet mismatch under the specified `/64` prefix lengths. Therefore, changing either address solely to fix a presumed prefix mismatch is not justified.

Other possible causes of communication failure include:

•Incorrect IPv6 configuration on the network interfaces.
•A disconnected link or incorrect VLAN assignment.
•Duplicate IPv6 addresses.
•IPv6 routing or Neighbor Discovery problems.
•Host firewall rules blocking traffic.
•Incorrect router configuration if the devices are on different Layer 2 segments.

These possibilities should be tested before modifying valid addresses.

5. Optimized configuration

The existing addresses can be retained, provided they are unique and configured correctly.

Parameter	Device A	Device B
IPv6 address	`2001:db8::ff00:42:8329/64`	`2001:db8:0:0:1::1/64`
Network prefix	`2001:db8:0:0::/64`	`2001:db8:0:0::/64`
Suggested action	Verify interface and connectivity	Verify interface and connectivity

The `2001:db8::/32` block is reserved for documentation, so these addresses are suitable for lab examples and simulations rather than direct public Internet communication.

6. Calculate available host addresses

A `/64` IPv6 subnet contains \(128-64=64\) interface-identifier bits. Interface identifiers=2^(64)

2^(64)=18,446,744,073,709,551,616

Thus, the subnet contains 2^(64)possible interface identifiers. This is not necessarily the number of addresses that can be assigned under every IPv6 addressing rule.

Final conclusion: Both IPv6 addresses have the same `/64` network prefix. No prefix mismatch exists in the supplied configuration. The correct debugging approach is to preserve the valid addresses and investigate Layer 2 connectivity, Neighbor Discovery, routing, duplicate addresses, and firewall settings.

Q3. IPv4 Subnet Allocation and VLSM Optimization

1. Problem identification

The organization has allocated `192.168.1.0/24` to several departments using `/26`, `/27`, and `/28` subnet masks.

The reported problems are:

•Some departments do not have enough usable IP addresses.
•Other departments have many unused addresses.
•The addressing plan may contain overlapping subnets, incorrect network boundaries, or inefficient allocation.

The solution is to examine the address ranges and use Variable Length Subnet Masking (VLSM), which allocates subnets according to each department's actual host requirements.

2. Calculate the subnet details

The network `192.168.1.0/24` contains 256 total addresses, from `192.168.1.0` through `192.168.1.255`.

The following table shows the properties of each subnet size.

Prefix	Subnet mask	Total addresses	Usable IPv4 hosts	Block size
`/26`	`255.255.255.192`	64	62	64
`/27`	`255.255.255.224`	32	30	32
`/28`	`255.255.255.240`	16	14	16

The usable-host calculation for conventional IPv4 subnets is:

Usable hosts=2^(32-prefix)-2

The two excluded addresses are the network address and broadcast address.

3. Network addresses, broadcast addresses, and host ranges

The actual network boundaries depend on where each subnet is allocated within the `/24` block. The following is an example of a valid, non-overlapping allocation for one subnet of each size.

Subnet	Network address	Valid host range	Broadcast address
`/26`	`192.168.1.0`	`192.168.1.1–192.168.1.62`	`192.168.1.63`
`/27`	`192.168.1.64`	`192.168.1.65–192.168.1.94`	`192.168.1.95`
`/28`	`192.168.1.96`	`192.168.1.97–192.168.1.110`	`192.168.1.111`

The remaining range, `192.168.1.112–192.168.1.255`, is available for additional subnets, provided each new subnet uses a valid boundary and does not overlap an existing allocation.

This table is an example, not a reconstruction of the organization's actual allocation: the question does not provide the original departmental network addresses or host counts.

4.Calculate usable and unused IP addresses

The total usable-host capacity of one subnet of each size is:

62+30+14=106 usable hosts The three subnets occupy:

64+32+16=112 total addresses

The number of unused usable host addresses depends on the number of devices actually required by each department.

For a subnet with H required hosts:

Unused usable addresses =Usable host capacity-H For example, suppose a department requires

20 hosts.

- With a `/26`, it has 62 usable addresses and 42 unused usable addresses.
- With a `/27`, it has 30 usable addresses and 10 unused usable addresses.
4. A `/28` is insufficient because it supports only 14 usable hosts.

The `/27` is the smallest of these three subnet sizes that can accommodate 20 hosts, assuming no additional addresses are needed for future growth or special infrastructure.

5. Identify inefficient allocations and subnetting errors

Common problems include:

1. Oversized subnet: Allocating `/26` to a department that needs only 10 hosts wastes address capacity.
2. Undersized subnet: Allocating `/28` to a department requiring 20 hosts causes an address shortage.
5. Overlapping ranges: Assigning two departments subnets that cover the same addresses causes conflicts.
6. Invalid network boundary: A `/27` network must begin at a multiple of 32 in the last octet; a `/28` must begin at a multiple of 16.

- Incorrect broadcast address: Using an address outside the allocated block can cause configuration errors.

These issues should be checked against the actual subnet allocation table and the number of hosts required by each department.

1. Recommend an optimized VLSM scheme

VLSM allocates the largest required subnet first, then assigns smaller subnets from the remaining address space.

The following example assumes that three departments require 50, 20, and 10 host addresses respectively. These are illustrative requirements because the original question does not specify departmental host counts.

Example VLSM allocation

Department A — 50 hosts

/26

Network: `192.168.1.0/26`

Host range: `192.168.1.1–192.168.1.62`
Broadcast: `192.168.1.63` · 62 usable addresses Department B — 20 hosts

/27

Network: `192.168.1.64/27`

Host range: `192.168.1.65–192.168.1.94`
Broadcast: `192.168.1.95` · 30 usable addresses Department C — 10 hosts

/28

Network: `192.168.1.96/28`

Host range: `192.168.1.97–192.168.1.110`

Broadcast: `192.168.1.111` · 14 usable addresses

In this example, the three departments need 80 hosts and receive 106 usable addresses in total, leaving 26 usable host addresses beyond their stated requirements. The `/26`, `/27`, and `/28` allocations occupy 112 of the 256 addresses in the original `/24` network.

Final conclusion: VLSM improves IPv4 address utilization by matching subnet size to host demand. To produce an exact optimized plan for the organization, the actual host requirement of each department and the existing subnet boundaries must be known.

Q4. IPv6 Address Debugging and Optimization

1. Problem identification

Given:

•Device A: `2001:db8::ff00:42:8329/64`
•Device B: `2001:db8:0:0:1::1/64`

The objective is to identify the network prefix, verify subnet membership, and locate the actual cause of communication failure.

2. Expand the addresses

Device	Complete IPv6 address
A	`2001:0db8:0000:0000:0000:ff00:0042:8329/64`
B	`2001:0db8:0000:0000:0001:0000:0000:0001/64`

Each expanded address has eight 16-bit groups, giving a total length of 128 bits.

3.Determine subnet membership

For both devices, the first four groups form the network prefix:

- Device A: `2001:db8:0:0::/64`
- Device B: `2001:db8:0:0::/64`

Therefore, both devices belong to the same IPv6 subnet.

The address `2001:db8:0:0:1::1` does not indicate a different `/64` subnet. Its fifth group is part of the interface identifier.

3.Debugging procedure

The supplied configurations do not establish a prefix mismatch. Use the following sequence to locate the failure.

•Check that both IPv6 addresses and `/64` prefix lengths are configured on the correct interfaces.
•Verify that both interfaces are up and connected to the correct VLAN or Layer 2 network.
•Check that the addresses are unique.
•Inspect IPv6 Neighbor Discovery to determine whether each device can resolve the other's link-layer address.
•Check host firewalls and access-control rules.
•If the devices are separated by routers, verify the IPv6 routes and router interface configurations.

5. Optimized configuration

Keep the original addresses if they are correctly assigned and unique:

Setting	Device A	Device B
IPv6 address	`2001:db8::ff00:42:8329/64`	`2001:db8:0:0:1::1/64`
Network prefix	`2001:db8:0:0::/64`	`2001:db8:0:0::/64`
Correction required	Diagnose connectivity	Diagnose connectivity

No address change is necessary merely to align the network prefixes.

6. Calculate available addresses

The number of possible 64-bit interface identifiers in a `/64` subnet is: 2^(128-64)=2^64)

=18,446,744,073,709,551,616

Final conclusion: Both devices share the same `/64` prefix, so the given addresses are subnet-compatible. The correct optimization is to verify Layer 2 connectivity, Neighbor Discovery, routing where applicable, and firewall configuration rather than introducing an unnecessary addressing change.

Q5. Debugging and Optimizing a Router's Routing Table

1. Problem identification

The router contains the following routing table.

Destination network	Next hop
`192.168.1.0/24`	`10.0.0.2`
`192.168.2.0/24`	`10.0.0.3`
Default route (`0.0.0.0/0`)	`10.0.0.1`

Users report that packets destined for `192.168.2.100` are not reaching the intended network.

The problem must be analyzed using longest-prefix matching, next-hop reachability, and route verification.

2.Identify the matching routing entry

The destination IP address is:

`192.168.2.100`
The route `192.168.2.0/24` covers the address range `192.168.2.0–192.168.2.255`.

Therefore, `192.168.2.100` belongs to the `192.168.2.0/24` subnet.

The router has two possible matching routes to consider:

2. `192.168.2.0/24` — a specific route.
3. `0.0.0.0/0` — the default route, which matches all IPv4 destinations.

The `/24` route has a longer prefix than `/0`, so it is selected.

3. Determine the next-hop router

The selected route is:

Parameter	Result
Destination IP	`192.168.2.100`
Matching network	`192.168.2.0/24`
Prefix length	`/24`
Selected next hop	`10.0.0.3`
Default route used?	No

Explanation: The router forwards the packet toward `10.0.0.3`, provided that the route is installed, the next hop is reachable, and no higher-priority forwarding condition changes the outcome.

4.Debugging and root-cause analysis

Since the destination matches the configured `/24` route, the routing table's longest-prefix decision is correct.

If packets still fail to reach the destination, investigate these possible causes:

- The next-hop address `10.0.0.3` is unreachable.
- The route is configured but not installed in the active routing table.
- The next-hop router lacks a route to the destination network.
- The destination network's subnet mask or host configuration is incorrect.
- An interface is down or a VLAN/link configuration is wrong.
- An access-control list or firewall is dropping the packets.
- The destination host's return route is missing, causing asymmetric or failed communication.

The supplied routing table alone does not prove which of these is the actual fault. The next-hop connectivity and the return path must be tested.

4.Optimize the routing table

The existing table already contains a specific route for `192.168.2.0/24` and a default route. No additional route is required for `192.168.2.100` if the existing entry is correct.

A suitable routing table is:

Destination	Next hop	Purpose
`192.168.1.0/24`	`10.0.0.2`	Reach network 1
`192.168.2.0/24`	`10.0.0.3`	Reach network 2
`0.0.0.0/0`	`10.0.0.1`	Reach destinations without a more specific route

Optimization should focus on verifying next-hop reachability, removing genuinely obsolete or conflicting routes, and ensuring the downstream routers have correct forward and return paths. The default route should remain if `10.0.0.1` is the intended upstream router.

6. Identify the route used when no specific destination matches

The default route is:

`0.0.0.0/0 → 10.0.0.1`

If the destination does not match either `192.168.1.0/24` or `192.168.2.0/24`, the router selects the default route, assuming it is present and usable in the active routing table.

For example, a destination of `8.8.8.8` would match neither specific `/24` route, so the router would use the default next hop `10.0.0.1`.

7.Verification using Cisco Packet Tracer

The following commands can be used on a Cisco IOS router to verify the routing behavior.

```
show ip route
show ip interface brief ping 10.0.0.3
traceroute 192.168.2.100
```

- `show ip route` displays installed routes and their next hops.
7. `show ip interface brief` checks whether router interfaces are operational.
- `ping 10.0.0.3` tests reachability to the intended next hop.
- `traceroute 192.168.2.100` helps identify where packets stop along the path.

Final conclusion: The destination `192.168.2.100` matches `192.168.2.0/24`, and the correct next hop is `10.0.0.3`. If that route is usable, the default route is not selected. The next-hop router, downstream routing, destination configuration, and return path should be verified to identify the actual communication failure

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understandin g	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justificatio n	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentatio n	Code is mostly clear with minor documentat ion gaps	Code lacks structure and proper documentatio n	Poorly written code with no documentatio n